CC LAB 8

SNEHA VERMA

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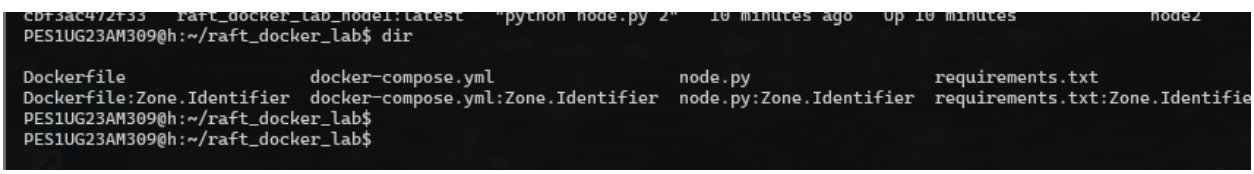
Aim and Objective

The primary **aim** of this project is to implement and demonstrate a distributed fault-tolerant system using the **Raft Consensus Algorithm** within a containerized environment.

The specific **objectives** are:

- **Orchestration:** To use Docker Compose to manage a multi-node cluster with isolated networking and static IP assignments.
- **Leader Election:** To observe how nodes transition between states (Follower, Candidate, and Leader) based on network availability and heartbeat signals.
- **Fault Tolerance:** To understand how the cluster maintains consistency even when individual nodes fail or become unreachable.
- **Distributed State:** To utilize the raftos library to synchronize state across multiple independent Python instances.

SS 1-



SS 2-

```
PES1UG23AM309@h:~$ cd raft_docker_lab
PES1UG23AM309@h:~/raft_docker_lab$ docke
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.2:800
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
```

SS 3-

```
PES1UG23AM309@h:/mnt/c/Users/ASUS$ cd ~
PES1UG23AM309@h:~$ cd raft_docker_lab
PES1UG23AM309@h:~/raft_docker_lab$ docker logs -f n
Starting Node 2 at 172.28.0.3:8002
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
```

```
PES1UG23AM309@h:~/raft_docker_lab$ docker logs -f node3
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
```

SS4:

```
PES1UG23AM309@h:~/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                  COMMAND              CREATED              STATUS              PORTS     NAMES
57360ec4e658   raft_docker_lab_node1  "python node.py 1"   About a minute ago   Up About a minute             node1
4b8ecc6723fa   raft_docker_lab_node3  "python node.py 3"   About a minute ago   Up About a minute             node3
b4863db1965f   raft_docker_lab_node2  "python node.py 2"   About a minute ago   Up About a minute             node2
PES1UG23AM309@h:~/raft_docker_lab$
```

SS5:

```
PES1UG23AM309@h:~/raft_docker_lab$ docker network inspect raft_docker_lab_raftnet
[
    {
        "Name": "raft_docker_lab_raftnet",
        "Id": "ed3cbf79d40a0a65411102db02da7da57a8d141690380d7006e929a0a2811712",
        "Created": "2026-03-27T09:00:38.479808152Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.28.0.0/16"
                }
            ]
        },
        "Internal": false,
        "Attachable": true,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {
            "4b8ecc6723fa142b60fe38d76331c5103e1f6a662db6d6ea6716fe6bb0343513": {
                "Name": "node3",
                "EndpointID": "d0fd4dcdee5aef4ee35f43c9f5bda29382160dcb9df014515a84e645a0033fc1",
                "MacAddress": "e2:a1:e9:0e:38:62",
                "IPv4Address": "172.28.0.4/16",
                "IPv6Address": ""
```

```
Windows PowerShell    Windows PowerShell
                "MacAddress": "e2:a1:e9:0e:38:62",
                "IPv4Address": "172.28.0.4/16",
                "IPv6Address": ""
            },
            "57360ec4e658c26ce8ebd13a5a357fd6100f99e80e5cce358639974609507ea9": {
                "Name": "node1",
                "EndpointID": "4254b09dbd8d28dfded5143ef610046a686489d41fd71ec5095d90735275cb46",
                "MacAddress": "96:69:0e:a3:fc:72",
                "IPv4Address": "172.28.0.2/16",
                "IPv6Address": ""
            },
            "b4863db1965ff86e5c7c4f54216121de97a0f83799881bc3de283cb40e329ff1": {
                "Name": "node2",
                "EndpointID": "9d938f3a24ea966742461e2059299d0b752dba3e986b5829ab6065b391e177eb",
                "MacAddress": "be:81:2a:69:d7:80",
                "IPv4Address": "172.28.0.3/16",
                "IPv6Address": ""
            }
        },
        "Options": {},
        "Labels": {
            "com.docker.compose.network": "raftnet",
            "com.docker.compose.project": "raft_docker_lab",
            "com.docker.compose.version": "1.29.2"
        }
    }
]
```

SS 6-

```
PES1UG23AM309@h:~/raft_docker_lab$ docker stop node1
node1
PES1UG23AM309@h:~/raft_docker_lab$
```

```
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
PES1UG23AM309@h:~/raft_docker_lab$
```

```
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[Node 2] Leader changed: 172.28.0.3:8002
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
```

```
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[Node 3] Leader changed: None
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

```
node1     | [LEADER Node 1] I am the leader
node1 exited with code 137
node2     | [FOLLOWER Node 2] Current leader: 172.28.0.2:8001
node3     | [Node 3] Leader changed: None
node3     | [Node 3] Waiting for leader election...
node2     | [Node 2] Leader changed: 172.28.0.3:8002
node2     | [LEADER Node 2] I am the leader
node3     | [Node 3] Leader changed: 172.28.0.3:8002
node3     | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

SS 7-

```
PES1UG23AM309@h:~/raft_docker_lab$ docker start node1
node1
```

```
node1     | [Node 1] Waiting for leader election...
node2     | [LEADER Node 2] I am the leader
node3     | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node1     | [Node 1] Leader changed: 172.28.0.3:8002
node1     | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2     | [LEADER Node 2] I am the leader
node3     | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node1     | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2     | [LEADER Node 2] I am the leader
node3     | [FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node1     | [FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2     | [LEADER Node 2] I am the leader
```

```
[Node 2] Leader changed: 172.28.0.3:8002
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
[LEADER Node 2] I am the leader
```

```
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

SS 8-

```
PES1UG23AM309@h:~/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                   COMMAND              CREATED          STATUS          PORTS     NAMES
5f350ac9261a   raft_docker_lab_node1   "python node.py 1"   24 seconds ago   Up 23 seconds             node1
866e3d470d84   raft_docker_lab_node3   "python node.py 3"   24 seconds ago   Up 23 seconds             node3
34260d50c830   raft_docker_lab_node4   "python node.py 4"   24 seconds ago   Up 23 seconds             node4
d75a5815fece   raft_docker_lab_node5   "python node.py 5"   24 seconds ago   Up 23 seconds             node5
93f10d4ffaf4   raft_docker_lab_node2   "python node.py 2"   24 seconds ago   Up 23 seconds             node2
PES1UG23AM309@h:~/raft_docker_lab$
```

SS 9:

```
PES1UG23AM309@h:~/raft_docker_lab$ docker logs -f node1
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.2:8001
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
```

SS 10:

```
PES1UG23AM309@h:~/raft_docker_lab$ docker logs -f node2
Starting Node 2 at 172.28.0.3:8002
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
```

SS 11:

```
PES1UG23AM309@h:~/raft_docker_lab$ docker logs -f node3
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
```

SS 12-

```
PES1UG23AM309@h:~/raft_docker_lab$ docker logs -f node4
Starting Node 4 at 172.28.0.5:8004
[Node 4] Waiting for leader election...
[Node 4] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
[FOLLOWER Node 4] Current leader: 172.28.0.2:8001
```

SS 13:

```
PES1UG23AM309@h:~$ cd raft_docker_lab
PES1UG23AM309@h:~/raft_docker_lab$ docker logs -f node5
Starting Node 5 at 172.28.0.6:8005
[Node 5] Waiting for leader election...
[Node 5] Leader changed: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
[FOLLOWER Node 5] Current leader: 172.28.0.2:8001
```

SS 14:

```
PES1UG23AM309@h:~/raft_docker_lab$ docker stop node4
node4
PES1UG23AM309@h:~/raft_docker_lab$ docker stop node5
node5
PES1UG23AM309@h:~/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                          COMMAND              CREATED         STATUS         PORTS     NAMES
592ac9752a31   raft_docker_lab_node1:latest   "python node.py 1"   6 minutes ago   Up 6 minutes             node1
cbf3ac472f33   raft_docker_lab_node1:latest   "python node.py 2"   6 minutes ago   Up 6 minutes             node2
52057f540f1d   raft_docker_lab_node1:latest   "python node.py 3"   6 minutes ago   Up 6 minutes             node3
PES1UG23AM309@h:~/raft_docker_lab$
```

SS 15-

```
PES1UG23AM309@h:~/raft_docker_lab$ docker stop node3
node3
PES1UG23AM309@h:~/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                          COMMAND              CREATED         STATUS         PORTS     NAMES
592ac9752a31   raft_docker_lab_node1:latest   "python node.py 1"   7 minutes ago   Up 7 minutes             node1
cbf3ac472f33   raft_docker_lab_node1:latest   "python node.py 2"   7 minutes ago   Up 7 minutes             node2
PES1UG23AM309@h:~/raft_docker_lab$
```

SS 16-

```
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[LEADER Node 1] I am the leader
[Node 1] Leader changed: None
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
```

SS 17:

```
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[Node 2] Leader changed: None
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
```

Why Leader Election Failed with Only 2 Nodes

In a 5-node cluster, a leader election cannot occur if only 2 nodes are active because Raft requires a **Majority Quorum** to elect a leader.

1. The Majority Rule

The fundamental rule of Raft is that a candidate must receive votes from a majority of the cluster to become a leader. The formula for a majority is:

$$\text{Quorum} = \left\lfloor \frac{N}{2} \right\rfloor + 1$$

For your 5-node cluster ($N=5$):

$$\left\lfloor \frac{5}{2} \right\rfloor + 1 = 3$$

2. Safety and Split Brains

Raft enforces this majority requirement to prevent a "Split Brain" scenario. If a leader could be elected with only 2 nodes in a 5-node cluster, you could theoretically have two different groups of 2 nodes both electing their own leaders simultaneously (leaving 1 node isolated). By requiring 3 votes, Raft ensures that only **one** leader can exist at any given time.

3. The Current State

With only 2 nodes active:

- A node will time out and become a **Candidate**.
- It votes for itself (1 vote).
- It requests a vote from the only other active node (potentially getting a 2nd vote).
- Because $2 < 3$, the candidate never reaches the threshold. It will continue to time out and restart the election indefinitely until a 3rd node joins the network.